VIT CHENNAI

Department of Mathematics, School of Advanced Sciences

organizes a

Faculty Development Program on Data Structures and Deep Learning for Artificial Intelligence

March 14, 16, 18–20, 2026

Program Schedule

Venue
AB2 – 301A

Day 1 — Basic Data Structures & Arrays (March 14, 2026)

Time	Session	TH (mins)	LAB (mins)	Speaker
09:00 – 09:30	**Inauguration**			
09:30 – 11:00	**Arrays & Linear Data Structures** Array operations — dynamic arrays — time complexity analysis *Lab: Array-based search and sort*	60	30	**Dr. Umitty Srinivasa Rao** SCOPE, VIT Chennai
11:00 – 11:15	**Tea Break**			
11:15 – 12:30	**Queues & Stacks** Queue operations — circular queue — stack operations *Lab: Queue and stack implementation*	45	30	**Dr. Umitty Srinivasa Rao** SCOPE, VIT Chennai
12:30 – 14:00	**Lunch Break**			
14:00 – 15:15	**Linked Lists** Singly and doubly linked lists — traversal and manipulation *Lab: Implement linked lists*	45	30	**Dr. Umitty Srinivasa Rao** SCOPE, VIT Chennai
15:15 – 15:30	**Tea Break**			
15:30 – 17:00	**Day 1 Assessment** DSA coding problems — time complexity analysis	—	90	**Dr. Umitty Srinivasa Rao** SCOPE, VIT Chennai

Day 2 — Advanced Data Structures (March 16, 2026)

Time	Session	TH (mins)	LAB (mins)	Speaker
09:00 – 11:00	**Supervised Learning** Classification — regression — decision trees — k-NN — SVM *Lab: scikit-learn hands-on*	90	30	**Dr. Arun Rajkumar** IIT Madras
11:00 – 11:15	**Tea Break**			
11:15 – 12:30	**Unsupervised Learning** Clustering — K-Means — PCA — dimensionality reduction *Lab: scikit-learn hands-on*	45	30	**Dr. Arun Rajkumar** IIT Madras
12:30 – 14:00	**Lunch Break**			
14:00 – 15:15	**Trees and Binary Search Trees** Tree traversals — BST operations — insertion and deletion *Lab: BST insert and delete*	45	30	**Dr. Kannadasan K** IIITDM, Kanchipuram
15:15 – 15:30	**Tea Break**			
15:30 – 17:00	**Advanced Trees** AVL trees — B-trees — Red-Black trees — 2-3 and 2-3-4 trees — B+ trees *Lab: Implementation of B-trees*	75	30	**Dr. Kannadasan K** IIITDM, Kanchipuram

Day 3 — Machine Learning & Neural Networks (March 18, 2026)

Time	Session	TH (mins)	LAB (mins)	Speaker
09:00 – 11:00	**Hash Tables and Hashing** Hash functions — collision resolution strategies — open addressing vs. chaining *Lab: Hash table CRUD operations*	45	30	**Dr. Jayaram B** SCOPE, VIT Chennai
11:00 – 11:15	**Tea Break**			
11:15 – 12:30	**Heaps and Priority Queues** Min/max heaps — heap sort — priority queues using arrays, linked lists, and heaps *Lab: Heap and priority queue implementation*	45	30	**Dr. Jayaram B** SCOPE, VIT Chennai
12:30 – 14:00	**Lunch Break**			
14:00 – 15:15	**Neural Network Fundamentals** Perceptrons — activation functions — forward pass — loss functions *Lab: Build a simple neural network in PyTorch/TensorFlow*	45	30	**Dr. Jeganathan L** SCOPE, VIT Chennai
15:15 – 15:30	**Tea Break**			

Time	Session	TH (mins)	LAB (mins)	Speaker
15:30 – 17:00	**Backpropagation Algorithm** Gradient descent — chain rule — weight update mechanics — vanishing gradients *Lab: Manual backpropagation*	60	30	**Dr. Jeganathan L** SCOPE, VIT Chennai

Day 4 — Deep Learning Architectures (March 19, 2026)

Time	Session	TH (mins)	LAB (mins)	Speaker
09:00 – 11:00	**Multi-Layer Perceptrons** Hidden layers — universal approximation theorem — depth vs. width trade-offs *Lab: MLP on the XOR problem*	90	30	**Dr. Jeganathan L** SCOPE, VIT Chennai
11:00 – 11:15	**Tea Break**			
11:15 – 12:30	**Optimizers and Loss Functions** SGD — momentum — Adam — cross-entropy — regularization techniques (dropout, L2) *Lab: Optimizer comparison*	45	30	**Dr. Jeganathan L** SCOPE, VIT Chennai
12:30 – 14:00	**Lunch Break**			
14:00 – 15:15	**Recurrent Neural Networks** Vanilla RNN — LSTM — GRU — sequence modeling — vanishing gradient problem *Lab: Text classification*	45	30	**Dr. Janaki Meena** SCOPE, VIT Chennai
15:15 – 15:30	**Tea Break**			
15:30 – 17:00	**Attention Mechanisms and Transformer Architecture** Self-attention — multi-head attention — encoder-decoder — positional encoding *Lab: Mini-transformer implementation*	60	30	**Dr. Janaki Meena** SCOPE, VIT Chennai

Day 5 — Advanced Deep Learning & DSA Integration (March 20, 2026)

Time	Session	TH (mins)	LAB (mins)	Speaker
09:00 – 11:00	**Convolutional Neural Networks** Convolution layers — pooling — feature maps — stride and padding *Lab: CNN on MNIST*	75	45	**Dr. Saraswathi D** SCOPE, VIT Chennai
11:00 – 11:15	**Tea Break**			
11:15 – 12:30	**Advanced CNN Architectures** ResNet — VGG — Inception — transfer learning and fine-tuning *Lab: Fine-tuning ImageNet pre-trained models*	45	30	**Dr. Saraswathi D** SCOPE, VIT Chennai
12:30 – 14:00	**Lunch Break**			
14:00 – 15:15	**Object Detection and Image Segmentation** YOLO — Faster R-CNN — anchor boxes Semantic vs. instance segmentation — U-Net — Mask R-CNN *Lab: Detection and segmentation demo*	45	30	**Dr. Saraswathi D** SCOPE, VIT Chennai
15:15 – 15:30	**Tea Break**			
15:15 – 16:30	**Generative Models and Vision Transformers** GANs — image synthesis — variational autoencoders (VAEs) Attention maps — Vision Transformers (ViT) *Lab: GAN image generation*	45	30	**Dr. Saraswathi D** SCOPE, VIT Chennai
16:30 – 17:00	**Valedictory Ceremony**			

Note

- Each session includes a dedicated hands-on lab component. Participants are kindly requested to bring their own laptops.
- Lab hours shown in the schedule are indicative only and may vary at the discretion of the speaker.
- Sessions are integrated by design — speakers may move between theory and practice fluidly throughout.